Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The position x of a particle varies with time (t) as $x = at^2 - bt^3$. The acceleration at time t of the particle will be equal to zero, where t is equal to

(A) $\frac{2a}{3b}$

(B) $\frac{a}{b}$

(C) $\frac{a}{3b}$

(D) zero

Q.2

Two cars A and B are travelling in the same direction with velocities V_A and V_B $(V_A > V_B)$. When car A is at a distance s behind car B, the driver of car A applies the brakes producing a uniform retardation a; there will be no collision when

(A) $s < \frac{(V_A - V_B)^2}{2a}$

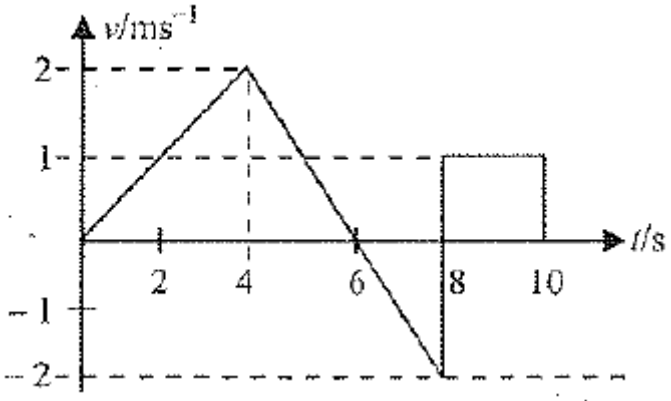
(B) $s = \frac{(V_A - V_B)^2}{2a}$

(C) $s \geq \frac{(V_A - V_B)^2}{2a}$

(D) $s \leq \frac{(V_A - V_B)^2}{2a}$

Q.3

The velocity-time graph of a body moving in a straight line is shown in Fig. The displacement of the body in 10 s is



(A) 4 m

(B) 6 m

(C) 8 m

(D) 10 m

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Q.4

Check up only the correct statement in the following.

(A) A body has a constant velocity and still it can have a varying speed

(B) A body has a constant speed but it can have a varying velocity.

(C) A body having constant speed cannot have any acceleration.

(D) None of these.

Q.5

Taxies leave the station X for station Y every $10\ \text{min}$. Simultaneously, a taxi also leaves the station Y for station X every $10\ \text{min}$. The taxies move at the same constant speed and go from X and Y or vice-versa in $2\ \text{h}$. How many taxies coming from the other side will meet each taxi enroute from Y and X?

(A) 24

(B) 23

(C) 12

(D) 11

Q.6

A person travels along a straight road for the first half time with a velocity v_1 and the second half time with a velocity v_2. Then the mean velocity $\bar{v}$ is given by

(A) $\bar{v} = \frac{v_1+v_2}{2}$

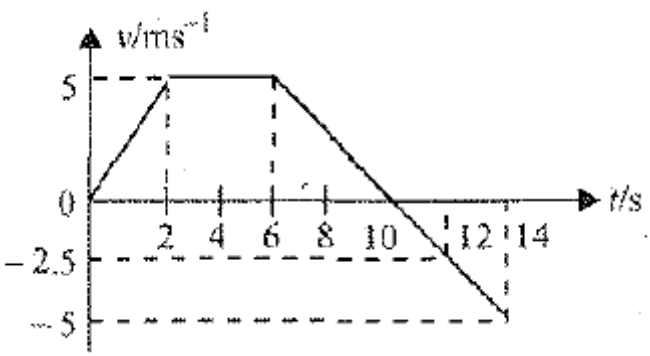
(B) $\frac{2}{\bar{v}} = \frac{1}{v_1} + \frac{1}{v_2}$

(C) $\bar{v} = \sqrt{v_1 v_2}$

(D) $\bar{v} = \sqrt{\frac{v_2}{v_1}}$

Q.7

The variation of velocity of a particle moving along a straight line is shown in Fig. The distance travelled by the particle in $12\ \text{s}$ is



(A) $37.5\ \text{m}$

(B) $32.5\ \text{m}$

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(C) $35.0\ \mathrm{m}$

(D) None of these.

Q.8

Two trains P and Q are moving along parallel tracks with same uniform speed of $20\ \mathrm{m\ s^{-1}}$. The driver of train P decides to overtake train Q and accelerates his train by $1\ \mathrm{m\ s^{-2}}$. After $50\ \mathrm{s}$, train P crosses the engine of train Q. Find out what was the distance between the two trains initially, provided the length of each train is $400\ \mathrm{m}$.

(A) 450m

(B) 400m

(C) 350m

(D) 250m

Q.9

A car starts moving rectilinearly, first with acceleration $\alpha = 5\ \mathrm{ms^{-2}}$ (the initial velocity is equal to zero), then uniformly, and finally, decelerating at the same rate α comes to a stop. The total time of motion equals $t = 25\ \mathrm{s}$. The average velocity during this time is equal to $< \nu >= 72\ \mathrm{km\ h^{-1}}$. How long does the car move uniformly?

(A) 20s

(B) 15s

(C) 25s

(D) 10s

Q.10

The magnitude of the displacement is equal to the distance covered in a given interval of time if the particle

(A) moves with constant acceleration along any path

(B) moves with constant speed

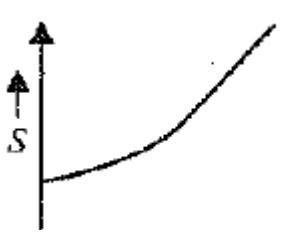
(C) moves in same direction with constant velocity or with variable velocity

(D) moves with constant velocity

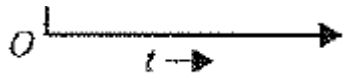
Q.11

Two balls are dropped from the top of a high tower with a time interval of t_o second, where t_o is smaller than the time taken by the first ball to reach the floor, which is perfectly inelastic. The distance s between the two balls, plotted against the time lapse t from the instant of dropping the second ball is best represented by

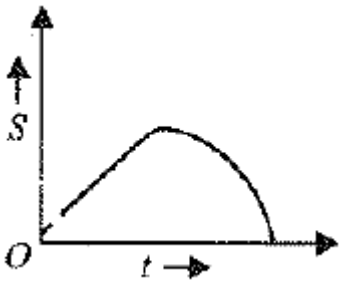
(A)



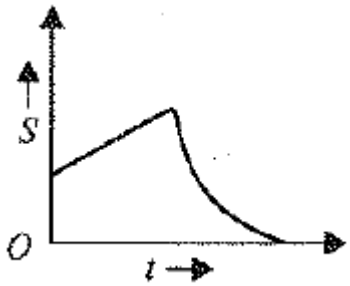
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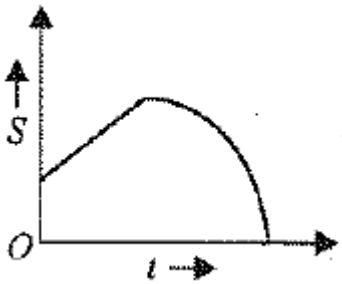
(B)



(C)



(D)



Q.12

A man loses 20% of his velocity after running through 108 m. Find maximum distance he can run. Further, if his retardation is uniform.

(A) 190m

(B) 185m

(C) 192m

(D) 200m

Q.13

Between two stations a train starting from rest first accelerates uniformly, then moves with constant velocity and finally retards uniformly to come to rest. If the ratio of the time taken be $1:8:1$ and the maximum speed attained be 60 km/h, then what is the average speed over the whole journey?

(A) 48 km/h

(B) 52 km/h

(C) 54 km/h

(D) 56 km/h

Q.14

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A particle starts moving rectilinearly at time $t = 0$ such that its Velocity v changes with time t according to the equation $v = t^2 - t$ where t is in seconds and v in ms^{-1}. Find the time interval for which the particle retards.

(A) $\frac{1}{2}s < t < 1s$

(B) $\frac{1}{3}s < t < \frac{s}{2}$

(C) $\frac{1}{4}s < t < 1s$

(D) $1s < t < \frac{3s}{2}$

Q.15

A body starting from rest moving with uniform acceleration has a displacement of $16\ \mathrm{m}$ in first $4\ \mathrm{s}$ and $9\ \mathrm{m}$ in first $3\ \mathrm{s}$. The acceleration of the body is

(A) $1\ \mathrm{ms}^{-2}$

(B) $2\ \mathrm{ms}^{-2}$

(C) $3\ \mathrm{ms}^{-2}$

(D) $4\ \mathrm{ms}^{-2}$

Q.16

A point moves in a straight line so that its displacement $x\ \mathrm{m}$ at time $t\ \mathrm{s}$ is given by $x^2 = 1 + t^2$. Its acceleration in $\mathrm{m/s}^2$ at lime $t\ \mathrm{s}$ is

(A) $\frac{1}{x^3}$

(B) $\frac{-t}{x^3}$

(C) $\frac{1}{x} - \frac{t^2}{x^3}$

(D) $\frac{1}{x} - \frac{1}{x^2}$

Q.17

The x and y coordinates of a particle at any time t are given by $x = 7t + 4t^2$ and $y = 5t$, where x and y are in m and t in s. The acceleration of the particle at $5\ \mathrm{s}$ is

(A) zero

(B) $8\ \mathrm{m/s}^2$

(C) $20\ \mathrm{m/s}^2$

(D) $40\ \mathrm{m/s}^2$

Q.18

A police party is chasing a dacoit in a jecp which is moving at a constant speed v. The dacoit is on a motor cycle. When he is at a distance x from the jeep he accelerates from rest at a constant rate. Which of the following relations

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is true, if the police is able to catch the dacoit?

(A) $v^2 \leq \alpha x$

(B) $v^2 \leq 2\alpha x$

(C) $v^2 \geq 2\alpha x$

(D) $v^2 \geq \alpha x$

Q.19

A $2\ \mathrm{m}$ wide truck is moving with a uniform speed $v_0 = 8\ \mathrm{m/s}$ along a straight horizontal road. A pedestrian starts to cross the road with a uniform speed v when the truck is $4\ \mathrm{m}$ away from him. The minimum value of v so that he can cross the road safely is

(A) $2,62\ \mathrm{m/s}$

(B) $4.6\ \mathrm{m/s}$

(C) $3.57\ \mathrm{m/s}$

(D) $1.414\ \mathrm{m/s}$

Q.20

A balloon is ascending vertically with an acceleration of 1 ms^{-2}. Two stones are dropped from it at an interval of $2\ \mathrm{s}$. Find the distance between them $1.5\ \mathrm{s}$ after the second stone is released.

(A) 45 m

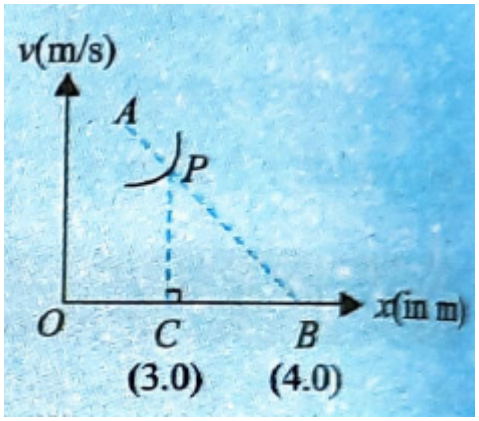
(B) 50 m

(C) 55 m

(D) 60 m

Q.21

A particle is moving along x axis and its velocity (v) vs position (x) graph is a curve as shown in the figure. Line APB is normal to the curve at point P. Find the instantaneous acceleration of the particle at $x = 3.0\ \mathrm{m}$.



Q.22

A parachutist drops first freely from an aeroplane for $10\ \mathrm{s}$ and then parachute opens out. Now he descends with a

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net retardation of $2.5\ \mathrm{m/s^2}$. If he bails out of the plane at a height of $2495\ \mathrm{m}$ and $g = 10\ \mathrm{m/s^2}$, his velocity on reaching the ground will be

Q.23

A stone is dropped from a certain height which can reach the ground in $5\ \mathrm{s}$. It is dropped after $3\ \mathrm{s}$ of its fall and then it again released. The total time taken by the stone to reach the ground will be

Q.24

A body sliding on a smooth inclined plane requires $4\ \mathrm{s}$ to reach the botom, starting from rest at the top. How much time does it take to cover one fourth the distance starting from rest at the top?

Q.25

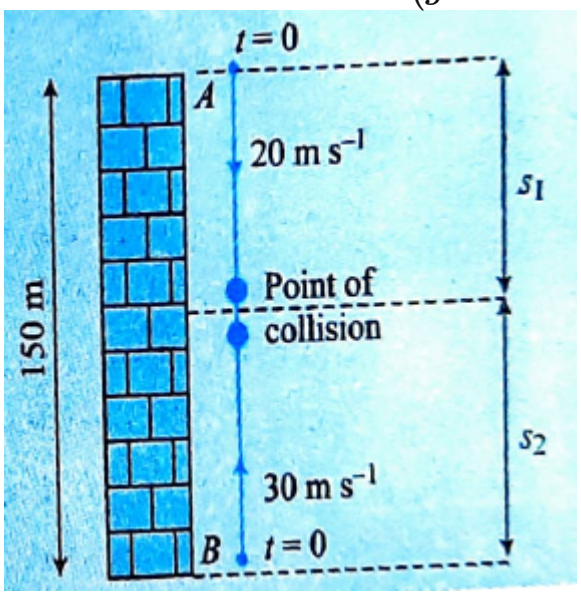
A moving car possesses average velocities of $5\ \mathrm{ms^{-1}}$, $10\ \mathrm{ms^{-1}}$ and $15\ \mathrm{ms^{-1}}$ in the first, second and third seconds, respectively. What is the total distance covered by the car in these 3 secs ?

Q.26

Water drops fall from a tap on the floor $5\ \mathrm{m}$ below at regular intervals of time, the first drop striking the floor when the fifh drop begins to fall. The height of the third drop from the ground, at the instant when the first drop strikes the ground, will be $\left(g = 10\ \mathrm{ms^{-2}}\right)$(mark answer to nearest integer of height)

Q.27

A ball is thrown downwards with a speed of $20\ \mathrm{ms^{-1}}$ from the top of a building $150\ \mathrm{m}$ high and simultaneously another ball is thrown vertically upwards with a speed of $30\ \mathrm{ms^{-1}}$ from the foot of the building. Find the time after which both the balls will meet. $\left(g = 10\ \mathrm{m\ s^{-2}}\right)$



Q.28

For motion of an object along x-axis, the velocity v depends on the displacement x as $v = 3x^2 - 2x$. What is the

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acceleration at $x = 2\ \text{m}$.What is the value of acceleration/10 ?

Q.29

A particle is projected up with initial speed $u = 10\ \text{m/s}$ from the top of a building at time $t = 0$. At time $t = 5\text{sec}$, the particle strikes the ground. Find the height/15 of the building.

Q.30

A stone is dropped from the 25th storey of a multistoried building and reaches the ground in $5\ \text{s}$. In the first second, it passes through how many storeys of the building? $(g = 10\ \text{m/s}^2)$

Answers & Solutions

Q.1 Answer:

$\frac{a}{3b}$

Solution:

$x = at^2 - bt^3$

$\text{velocity} = \frac{dx}{dt} = 2at - 3bt^2$

$\text{and acceleration} = \frac{d}{dt}\left(\frac{d^2x}{dt^2}\right) = 2a - 6bt$

Acceleration will be zero if

$2a - 6bt = 0 \Rightarrow t = \frac{2a}{6b} = \frac{a}{3b}$

Q.2 Answer:

$s \geq \frac{(V_A - V_B)^2}{2a}$

Solution:

For no collision, the speed of car A should be reduced to v_B before the cars meet, i.e. final relative velocity of car A with respect to car B is zero, i.e., $V_r = 0$

Here u_r = initial relative velocity $= V_A - V_B$

Relative acceleration $= a_r = -a - 0 = -a$

Let relative displacement $= s_r$

Then using the equation, $v_r^2 = u_r^2 + 2a_r s_r$

$0^2 = (V_A - V_B)^2 - 2as_r \cdot$ or $s_r = \frac{(V_A - V_B)^2}{2a}$

For no collision, $s_r \leq s$, i.e., $\frac{(V_A - V_B)^2}{2a} \leq s$

Q.3 Answer:

$6\ \text{m}$

Solution:

We know that area under $v - t$ graph is displacement.

Area from 0 to $6\ \text{s} = \frac{1}{2} \times 6 \times 2 = 6\ \text{m}$

Area from 6 to $8s = \frac{1}{2} \times 2 \times (-2) = -2m$

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Area from 8 to $10\ \mathrm{s} = 2 \times 1 = 2\ \mathrm{m}$
So Net displacement $= 6 - 2 + 2 = 6\ \mathrm{m}$

Q.4 Answer:

A body has a constant speed but it can have a varying velocity.

Solution:

When a body possesses constant velocity, then both its magnitude (i.e., speed) and direction must remain constant. On the other hand, if the speed of a body is constant, then its velocity may or may not remain constant. For example, in uniform circular motion, though the speed of body remains constant but velocity changes from point to point due to change in direction.
A body moving with a constant speed along a circular path constantly experiences a centripetal acceleration.

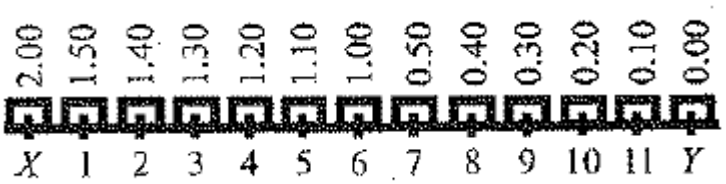
Q.5 Answer:

23

Solution:

We will first assume that 2 hours have passed since the first taxi started from X**. Let us consider one taxi starting from** Y **at this exact moment, at this moment of time there will be taxis at 10-minute intervals along the way. Now all the taxis that are in the way and started from** X **including the taxi that just reached** Y **and all the taxis that will start from** X **till the time this taxi from** Y **reaches** X **will cross this taxi from** Y **at some point of time. So, first we will calculate the number of taxis that are in the way.**

As all the taxis are 10 minutes apart and it takes 2 hours for a one-way trip, there will be 11 taxis on the way. The taxi that reached Y **and the taxi that will just begin at** X **are not counted as they are not enroute. After this taxi starts and takes 2 hours to reach** X**,** 12 **more taxis must have departed from** X **which will all cross this taxi from** Y**. So, in total 23 taxis coming from** X **will meet each taxi coming from** Y **and vice-versa.**



Alternate Solution:

We can also solve this problem using relative velocity. The relative velocity of each taxi with reference to a taxi coming from the other side. So, there are only 5-minute intervals in crossing each taxi. In the journey of 2 hours, there are 245 -minute intervals. But in the last interval no taxi will cross, so we get the same answer i.e. 23.

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Q.6 Answer:

$\bar{v} = \frac{v_1+v_2}{2}$

Solution:

$V_{\text{av}} = \frac{v_1(t/2)+v_2(t/2)}{t} = \frac{v_1+v_2}{2}$

Q.7 Answer:

37.5 m

Solution:

Area from 0 to $10s = \frac{1}{2}[10+4] \times 5 = 35 \text{ m}$

Area from 10 to $12 \text{ s} = \frac{1}{2} \times 2 \times (-2.5) = -2.5 \text{ m}$

Distance travelled $= 35 + 2.5 = 37.5 \text{ m}$

Q.8 Answer:

450m

Solution:

Let the initial distance between two trains be x**. Distance travelled by train** P **in** 50 s **:**

$S_P = ut + \frac{1}{2}at^2 = 20 \times 50 + \frac{1}{2} \times 1 \times 50^2 = 2250 \text{ m}$

Distance travelled by train Q **in** 50 s **:**

$S_Q = ut = 20 \times 50 = 1000 \text{ m}$

Now $S_P - S_Q = 2250 - 1000 = 1250 \text{ m}$**. This distance must be equal to the initial distance between the trains plus the sum of the lengths of the two trains.**

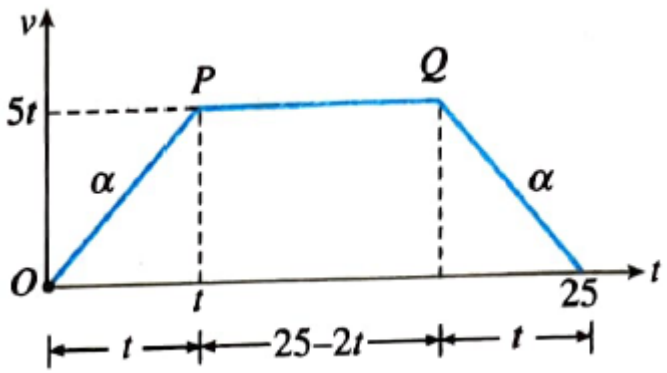
$x + 800 = 1250 \Rightarrow x = 450 \text{ m}$

Q.9 Answer:

15s

Solution:

Let t **be the time up to which the car accelerates or decelerates. The maximum velocity attained in this duration is** $5t$**. The time** u **pe which car moves uniformly** $= 25 - 2t$**. The velocity-time graph** $_{\text{of}}$ **the motion of car is drawn as shown in Figure.**



Given the average velocity is whole time of motion

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$v_{av} = \frac{72 \times 5}{18} = 20 \text{ ms}^{-1}$

The average velocity from the graph can be obtained as

$$v_{av} = \frac{\text{Total displacement}}{\text{Total time}} = \frac{\text{Area of } \vec{v}-t \text{ graph}}{\text{Total time}}$$

$$\Rightarrow \quad 20 = \frac{\frac{1}{2} \times [25 + (25 - 2t)] \times 5t}{25} = \frac{\frac{1}{2} \times [50 - 2t)] \times 5t}{25}$$

$$200 = 50t - 2t^2$$

$$t^2 - 25t + 100 = 0$$

$$\Rightarrow \quad (t - 20)(t - 5) = 0 \Rightarrow t = 5 \text{ s or } 20 \text{ s}$$

But $t = 20$ s is not possible.

Hence, $t = 5$ s

The time up to which car moves uniformly

$= 25 - 2t = 25 - 2 \times 5 = 15$ s

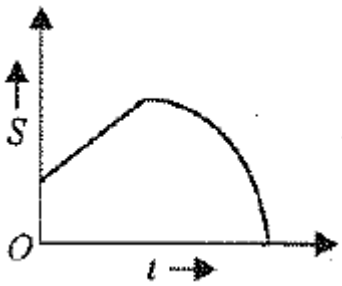
Q.10 Answer:

moves in same direction with constant velocity or with variable velocity

Solution:

To have distance equal to magnitude of displacement, the particle has to move in the same direction. The velocity may or may not be constant.

Q.11 Answer:



Solution:

Before the second ball is dropped, the first ball would have travelled some distance say $S_0 = \frac{1}{2}gt_0^2$. After dropping sccond ball, relative acceleration of both balls becomes zero. So distances between them increases linearly. After some time, the first ball will collide with the ground and the distance between them will start decreasing and magnitude of relative velocity will be increasing for this time. Option (d) represents all these clearly.

Q.12 Answer:

192m

Solution:

Let the initial velocity be u. Since the man loss 20% of his velocity after running through 108 m, Hence, velocity at this mos Now, initial velocity $= u$ Final velocity $= 0.8u$ and $s = 108$ m; $a = ?$ Final velocity $= 0.8u$ and $s = 108$ m; $a = ?$ Using $v^2 = u^2 + 2as \Rightarrow (0.8u)^2 = u^2 + 2a.(108)$

$216a = 0.64u^2 - u^2 = -0.36u^2 \Rightarrow a = -\frac{0.36}{216}u^2 = -\frac{1}{600}u^2$

Now, let the man run a further distance of x m. In second segment of his motion: Initial velocity $= 0.8u$, final velocity $= 0$; $s = x$; Again using $v^2 = u^2 + 2a.s \Rightarrow 0 = (0.8u)^2 + 2ax$ @IITJEE_Advanced

$$0 = 0.64u^2 + 2\left(-\frac{1}{600}u^2\right)x \Rightarrow \left(\frac{1}{300}u^2\right)x = 0.64u^2$$

$$\Rightarrow x = 0.64 \times 300 = 192 \text{ m}$$

Q.13 Answer:

54 km/h

Solution:

Given $u = 0$. Let during three phases time taken are t, $8t$. and t, respectively.

$$v_{\max} = at = 60 \text{ km/h}$$

$$v_{\text{av}} = \frac{\frac{1}{2}at^2 + v_{\max}8t + \frac{1}{2}at^2}{t + 8t + t} = \frac{at^2 + 8v_{\max}t}{10t} = \frac{at + 8v_{\max}}{10}$$

$$= \frac{60 + 8 \times 60}{10} = 54 \text{ km/h}$$

Q.14 Answer:

$\frac{1}{2}s < t < 1s$

Solution:

Acceleration of the particle, $\vec{a} = \frac{d\vec{v}}{dt} = 2t - 1$

The particle retards when acceleration is opposite to velocity. Hence, the acceleration vector and velocity vector should be opposite to each other or the dot product of $\vec{a}$ and $\vec{v}$ should be negative.

$$\Rightarrow \quad \vec{a} \cdot \vec{v} < 0$$

$$\Rightarrow \quad (2t - 1)\left(t^2 - t\right) < 0$$

t is always positive.

$$\therefore (2t - 1)(t - 1) < 0$$

$\Rightarrow$ Either $2t - 1 < 0$ or $t - 1 > 0$

$\Rightarrow t < \frac{1}{2}s$ and $t > 1s$. This is not possible. $\Rightarrow 2t - 1 > 0$ and $t - 1 < 0 \Rightarrow t > \frac{1}{2}s$ and $t < 1s$. **Hence, the required time interval is $\frac{1}{2}s < t < 1s$.**

Q.15 Answer:

2 ms^{-2}

Solution:

Distance travelled in 4th second $= 16 - 9 = 7$ m

Now $D_n = u + \frac{a}{2}(2n - 1)$,

Given $u = 0$

$\Rightarrow 7 = 0 + \frac{a}{2}(2 \times 4 - 1) \Rightarrow a = 2 \text{ m/s}^2$

Q.16 Answer:

$\frac{1}{x^3}$

Solution:

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Given $x^2 = 1 + t^2$

Differentiating the above equation we get

$2xdx = 2tdt$

$x\frac{dx}{dt} = t$ **...(1)**

Now taking differentiation for acceleration we get,

$x\frac{d^2x}{dt^2} + \left(\frac{dx}{dt}\right)^2 = 1$

$x\frac{d^2x}{dt^2} = 1 - \left(\frac{t}{x}\right)^2$

$x\frac{d^2x}{d^2} = \frac{x - \frac{t^2}{x}}{x} = \frac{(x^2 - t^2)}{x^3} = \frac{x^2 - t^2}{x^3}$

$\frac{d^2x}{dt^2} = \frac{x^2 - \left(x^2 + 1\right)}{x^3} = \frac{1}{x^3}$

Q.17 Answer:

$8\ \mathrm{m/s^2}$

Solution:

$a_x = \frac{d^2x}{dt^2} = 8$ **and** $a_y = \frac{d^2y}{dt^2} = 0$

Hence, net acceleration $= \sqrt{a_x^2 + a_y^2} = 8\ \mathrm{m/s^2}$

Q.18 Answer:

$v^2 \geq 2\alpha x$

Solution:

If police is able to catch the dacoit after time t**, then** $vt \geq x + \frac{1}{2}\alpha t^2$**. This gives** $\frac{\alpha}{2}t^2 - vt + x = 0$ **or** $t = \frac{v \pm \sqrt{v^2 - 2\alpha x}}{\alpha}$

For t **to be real,** $v^2 \geq 2\alpha x$

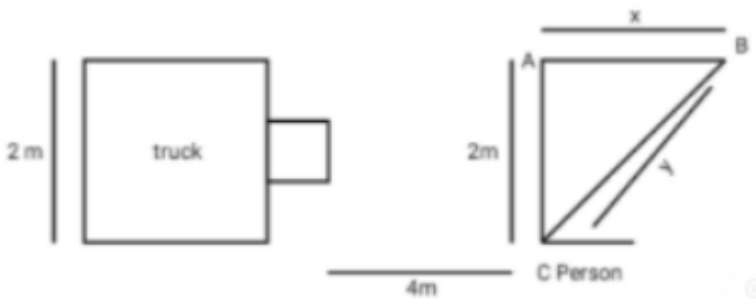
Q.19 Answer:

$3.57\ \mathrm{m/s}$

Solution:

A truck which is $2m$ **wide is moving with a uniform velocity along a straight line. At the same time a pedestrian starts to cross the road. Let us assume that person is in the position** C**, he wants to cross the road and reach point** B**. the distance between the truck and the person is** $4m$**. Let** A **be the point parallel to the person at** C **at a distance** $2m$**. With this draw a triangle joining the points** A**,** B **and** C**. Then** AC **is** $2m$**, let** AB **be** x **and let** BC **be y.**

Let us make a diagram using above discussion,



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The distance between the truck and the point B, $s = 4 + x$

The truck moves with the velocity, $v_T = 8ms^{-1}$

Let us take the triangle ABC**, it is a right angled triangle. According to the Pythagoras theorem,**

$\Rightarrow y = \sqrt{x^2 + 2^2}$

$y = \sqrt{x^2 + 4}$

The distance between the person at the point C **and the point** B **(other side),** $y = \sqrt{x^2 + 4}$

The person moves with the velocity $v_P = ?$

Velocity: the velocity is determined by the rate of change of displacement. I.e. displacement by time taken.

$\Rightarrow v = \frac{d}{t}$

This can be rewritten as

$\Rightarrow t = \frac{d}{v}$

V **is the velocity**

d is the displacement

t **is the time taken**

The time taken by the truck to reach the point B **is given by**

$\Rightarrow t = \frac{d}{v}$

$\Rightarrow t_T = \frac{4+x}{8} \rightarrow 1$

The time taken by the person to reach the point B **is given by**

$\Rightarrow t = \frac{d}{v}$

$\Rightarrow t_P = \frac{\sqrt{x^2+4}}{v_p} \rightarrow 2$

To cross the road safely without getting hit by the truck the time taken by the truck to reach the point B should be equal to the time taken by the person to reach the point B

So, equating 1 and 2

$\Rightarrow \frac{4+x}{8} = \frac{\sqrt{x^2+4}}{v_p}$

$\Rightarrow v_p = \frac{\sqrt{x^2+4}}{4+x} \times 8 \rightarrow 3$

To get minimum velocity of the person

$\Rightarrow \frac{dv}{dt} = 0$

$\Rightarrow \frac{d}{dt}\frac{\sqrt{x^2+4}}{4+x} \times 8 = 0$

$\Rightarrow 8\left(\frac{(4+x)\times\frac{x}{\sqrt{x^2+4}} - \sqrt{x^2+4}}{(x+4)^2}\right) = 0$

$\Rightarrow 8\left(\frac{(4+x)\times\frac{x-\left(x^2+4\right)}{\sqrt{x^2+4}}}{(x+4)^2}\right) = 0$

$\Rightarrow 8\left(\frac{(x+4)x-\left(x^2+4\right)}{\sqrt{x^2+4}}\right) = 0$

$\Rightarrow (x+4)x - \left(x^2+4\right) = 0$

$\Rightarrow x^2 + 4x - x^2 - 4 = 0$

$\Rightarrow 4x - 4 = 0$

$\Rightarrow 4(x-1) = 0$

$\Rightarrow (x-1) = 0$

$\Rightarrow x = 1$

Substitute the value of x **in equation 3**

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$\Rightarrow v_p = \frac{\sqrt{x^2+4}}{4+x} \times 8$

$\Rightarrow v_p = \frac{\sqrt{1^2+4}}{4+1} \times 8$

$\Rightarrow v_p = \frac{\sqrt{5}}{5} \times 8 = \frac{8}{\sqrt{5}}$

$\Rightarrow v_p = 3.57 \text{ ms}^{-1}$

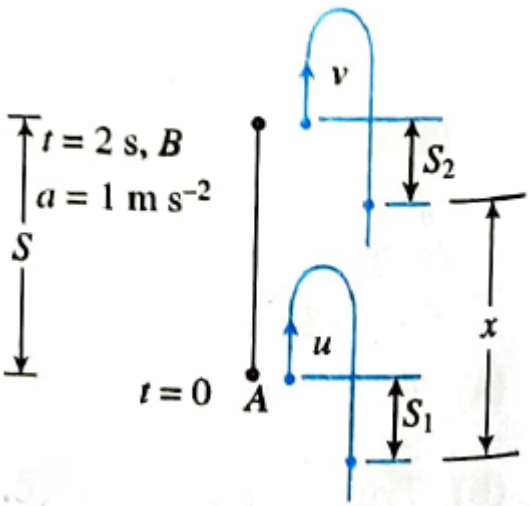
The minimum value of velocity required to cross the road safely is 3.57 ms^{-1}

Q.20 Answer:

55 m

Solution:

Let at any time $t = 0$**, the**



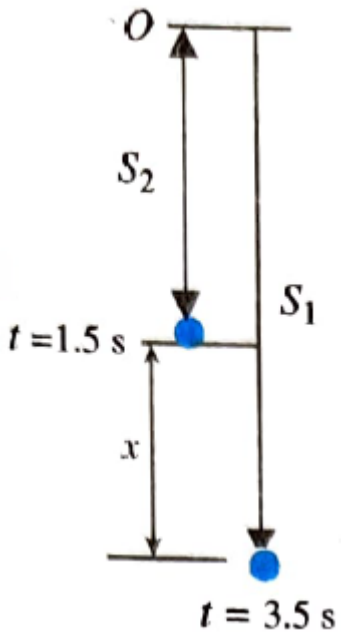
balloon be at position A**, where its velocity is** u**. At** $t = 2$ **s, it reaches** B, π **where its velocity becomes** v**, then**

$AB = S = u \times 2 + \frac{1}{2}a(2)^2 = 2u + 2$

Also $v = u + a \times 2 = u + 2$**,**

First stone: $-S_1 = u \times 3.5 + \frac{1}{2}(-g)(3.5)^2$

Second stone: $-S_2 = v \times 1.5 + \frac{1}{2}(-g)(1.5)^2$



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Required distance between the stones

$x = S_1 + S - S_2$

Solve to get $x = 55$ m.

Alternatively: (This method can be understood in proper way after studying relative velocity.) If we work from the frame of balloon, then the acceleration of each stone w.r.t. balloon will be $g + a$ **after releasing from it. The initial velocity of each stone will be zero w.r.t. balloon.**

$S_1 = \frac{1}{2}(g + a)(3.5)^2, S_2 = \frac{1}{2}(g + a)(1.5)^2; x = S_1 - S_2 = 55$ m

Q.21 Answer:

—

Solution:

Let the velocity of the particle at $x = 3.0$ m **be** v_0**. The slope of line** $APB = -\frac{v_0}{1}$

As the line APB **is normal to the curve at point** P**, hence the slope of tangent at** $P = \frac{1}{v_0}$

It means the slope of $v - x$ graph $\left(\frac{dv}{dx}\right)_P = \frac{1}{v_0}$

...(i)

But we define acceleration as $a = v\frac{dv}{dx}$ **...(ii)**

But at $x = 3.0$ m, $v = v_0$

and $\frac{dv}{dx} = \left(\frac{1}{v_0}\right) a = v_0\left(\frac{1}{v_0}\right) = 1$ m/s^2

Hence $a = v_0\left(\frac{1}{v_0}\right) = 1$ m/s^2

Q.22 Answer:

—

Solution:

The velocity v **acquired by the parachutist after** 10 s **:** $v = u + gt = 0 + 10 \times 10 = 100$ m/s

Then, $s_1 = ut + \frac{1}{2}gt^2 = 0 + \frac{1}{2} \times 10 \times 10^2 = 500$ m

The distance travelled by the parachutist under retardation, $s_2 = 2495 - 500 = 1995$ m

Let v_g **be this velocity on reaching the ground. Then** $v_g^2 - v^2 = 2as_2$

or $v_g^2 - (100)^2 = 2 \times (-2.5) \times 1995$ **or** $v_g = 5$ m/s

Q.23 Answer:

—

Solution:

Total height to fall

$h = \frac{1}{2}\text{gt}^2 = \frac{1}{2} \times 10 \times (5)^2 = 125m$

In first 3 sec.

$s_1 = \frac{1}{2}\text{gt}_1^2 = \frac{1}{2} \times 10(3)^2 = 45$ m

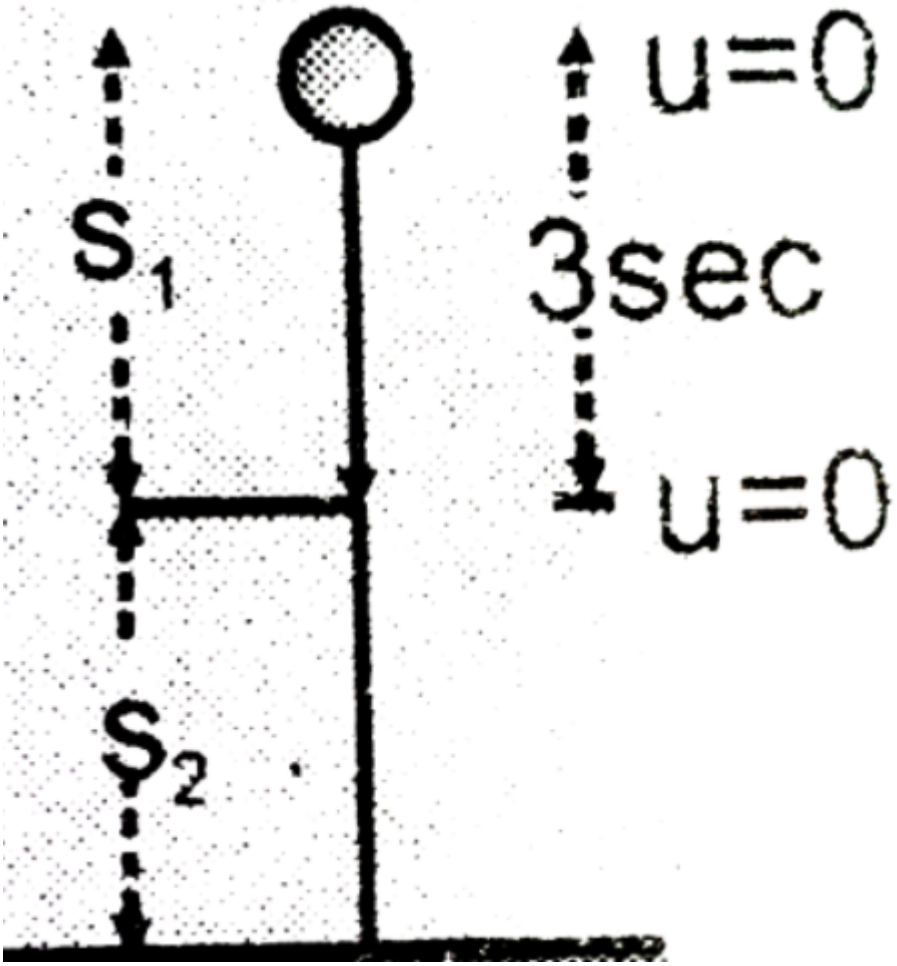
Now remaining height to fall

$\therefore s_2 = h - s_1 = 125 - 45 = 80m$

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$t_2 = \sqrt{\frac{2s_2}{g}} = \sqrt{\frac{2\times 80}{10}} = 4\text{sec}$

$t = t_1 + t_2 = 3 + 4 = 7\text{sec}$



Q.24 Answer:

—

Solution:

When a body slides on an inclined plane, component of weight along the plane produces an acceleration.

$a = \frac{mg\sin\hat{\theta}}{m} = g\sin\theta =$ **constant. If s be the length of the inclined plane, then**

$s = 0 + \frac{1}{2}at^2 = \frac{1}{2}g\sin\theta \times t^2$

$\frac{s'}{s} = \frac{t^2}{t^2}$ or $\frac{s}{s'} = \frac{t^2}{t'^2}$

Given, $t = 4s$ and $s' = \frac{s}{4}$

$t' = 1\sqrt{\frac{s'}{s}} = 4\sqrt{\frac{s}{4s}} = \frac{4}{2} = 2\text{ s}$

Q.25 Answer:

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Solution:

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distance covered $= s = v_{\text{av}} \times$ **time**

For first second: $S_1 = 5 \times 1 = 5 \text{ m}$

For second second: $S_2 = 10 \times 1 = 10 \text{ m}$

For third second: $S_3 = 15 \times 1 = 15 \text{ m}$

Total distance travelled

$S = S_1 + S_2 + S_3 = 5 + 10 + 15 = 30 \text{ m}$

Q.26 Answer:

—

Solution:

By the time 5 th water drop starts falling, the first water drop reaches the ground.

$u = 0, h = \frac{1}{2}gt^2 \Rightarrow 5 = \frac{1}{2} \times 10 \times t^2 \Rightarrow t = 1 \text{ s}$

Hence, the interval of falling of each water drop

$= \frac{1 \text{ s}}{4} = 0.25 \text{ s}$

When the 5 th drop starts its journey towards ground, the third drop travels in air for

$0.25 + 0.25 = 0.5 \text{ s}$

$\therefore$ Height (distance) covered by 3rd drop in air is

$h_1 = \frac{1}{2}gt^2 = \frac{1}{2} \times 10 \times (0.5)^2 = 5 \times 0.25 = 1.25 \text{ m}$

The third water drop will be at a height of

$= 5 - 1.25 = 3.75 \text{ m}$

Q.27 Answer:

—

Solution:

Method-1: Let the first ball move down distance S_1 and second ball moves up a distance S_2 before they meet. Let us take downward direction as positive.

Then $S_1 = 20t + 5t^2$

$S_2 = 30t - 5t^2$

But, $S_1 + S_2 = 150$

$\Rightarrow \quad 150 = 50t$

$\Rightarrow \quad t = 3 \text{ s}$

Method-2: Let us solve this problem by using the method of relative velocity.

Relative acceleration of both is zero since both have same acceleration in downward direction.

$\vec{a}_{AB} = \vec{a}_A = -\vec{a}_B = g - g = 0$

Initial relative velocity, $\vec{v}_{BA} = 30 - (-20) = 50 \text{ m s}^{-1}$

Relative separation between the particles $s_{B,A} = v_{B,A} \times t$

Hence, required time, $t = \frac{s_{BA}}{v_{Bd}} = \frac{150}{50} = 3 \text{ s}$

Q.28 Answer:

—

Solution:

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Given $v = 3x^2 - 2x$, differentiating v, we get

$\frac{dv}{dt} = (6x - 2)\frac{dx}{dt} = (6x - 2)v$

$\Rightarrow a = (6x - 2)\left(3x^2 - 2x\right)$. Now put $x = 2$ m

$\Rightarrow a = (6 \times 2 - 2)\left(3(2)^2 - 2 \times 2\right) = 80$ m/s^2

$\frac{\text{a}}{10} = \frac{80}{10}$

$= 8$ m/s^2

Q.29 Answer:

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Solution:

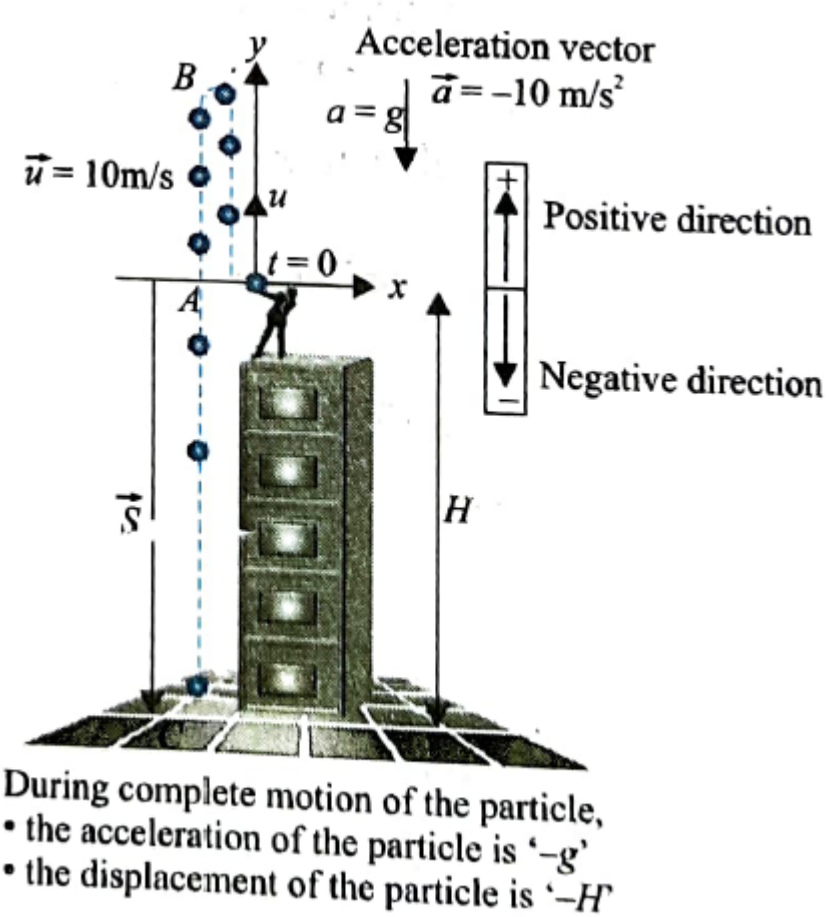
The particle is "freely falling" always. The expression "freely falling" does not necessarily mean an object is falling down A freely falling object is any object moving either upward downward under the influence of gravity alone.

The particle starts from A and finally reaches at C.

Let us take origin at A. Upward direction is taken as positive and downward direction is taken as negative.

The particle moves in gravitational field where acceleration due to the gravity is always acting in downward direction whether it is moving upward or downward.

Hence acceleration vector $\vec{a}$ will always be -10 m/s^2, ऽ its magnitude as well as direction remains constant always throughout the motion.



Hence acceleration, $\vec{a} = -10$ m/s^2

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Initially at $t = 0$, the particle is projected in upward direction. Hence, initial velocity $\vec{u} = 10\ \mathrm{m/s^2}$
The particle moves from A to B (upward) and then B to C (downward).
The motion of the particle goes from A to B and then again passes point A. The net displacement of the particle upto this instant is zero. Then particle crosses point A and finally reaches to C. We know that net displacement is equal to the difference of fol position vector.
Hence net displacement of the particle during motion $(t = 5\mathrm{sec})$
Using $\vec{s} = \vec{u}t + \frac{1}{2}\vec{a}t^2$
$\Rightarrow \quad \vec{s} = (10) \times 5 + \frac{1}{2}(-10)(5)^2 = 50 - 125 = 75(\ \mathrm{m})$
Hence $H = 75\ \mathrm{m}$

Q.30 Answer:

—

Solution:

Suppose h be the height of each storey, then
$25h = 0 + \frac{1}{2} \times 10 \times t^2 = \frac{1}{2} \times 10 \times 5^2$ or $h = 5\ \mathrm{m}$
In first second, let the stone passes through n storey, so $n \times 5 = \frac{1}{2} \times 10 \times (1)^2$ or $n = 1$

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